Linear functions — Part 1

Algebra · Math · 85 questions

1.1 fa992b50-1873-43ba-8b60-8c6d6bdc7bcc · Easy

The function g is defined by $gx = 6x$. For what value of x is $g(x) = 54$?

Student-produced response: _____

1.2 c31256e3-e43c-46a7-998b-9e8511243e7b · Easy

Sam has a flag with an area of $110 \text{ square inches } \text{in}^2$. Sam plans to enlarge the flag by sewing on pieces of colored nylon, where each piece would increase the flag's area by 64 in^2. Which equation gives the flag's total area y, in square inches, after Sam sews on x pieces of nylon?

A. $y = 64x$

B. $y = 64 + 110x$

C. $y = 110 + 64x$

D. $y = 174x$

1.3 df1e58e8-b763-42a1-a3e2-b70a4bac09d5 · Easy

Gabriella deposits $\$ 35$ in a savings account at the end of each week. At the beginning of the 1st week of a year there was $\$ 600$ in that savings account. How much money, in dollars, will be in the account at the end of the 4th week of that year?

A. 460

B. 635

C. 639

D. 740

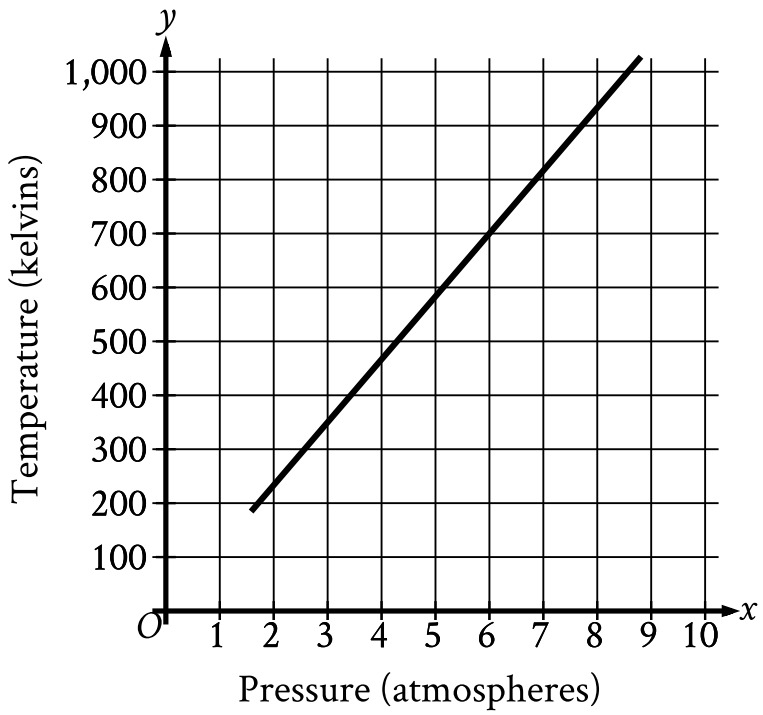
1.4 6aba0416-d25e-4a28-b655-c5623b3b71c0 · Easy

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

Student-produced response: _____

1.5 b347b50a-ec16-402a-b7e5-8ec5a188bf8f · Easy

Oxygen gas is placed inside a tank with a constant volume. The graph shows the estimated temperature y, in kelvins, of the oxygen gas when its pressure is x atmospheres.



- The line slants sharply up from left to right.
- The line begins at the approximate point (1.6 comma 190).
- The line passes through the following approximate points:
  - (3 comma 350)
  - (6 comma 700)
  - (8 comma 933)

What is the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres?

A. 6

B. 60

C. 700

D. 760

1.6 2050beca-5f4f-41a5-92ca-3fcf29a6cc77 · Easy

The number y is 84 less than the number x. Which equation represents the relationship between x and y?

A. $y = x + 84$

B. $y = \frac{1}{84}x$

C. $y = 84x$

D. $y = x - 84$

1.7 821016f4-912a-4737-9916-2b5d5ea434a3 · Easy

A contract for a certain service requires a onetime activation cost of $\$ 35$ and a monthly cost of $\$ 23$. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

A. $c = \frac{t}{23} + 35$

B. $c = \frac{t}{35} + 23$

C. $c = 23t + 35$

D. $c = 35t + 23$

1.8 029367-DC · Easy

Robert rented a truck to transport materials he purchased from a hardware store. He was charged an initial fee of $20.00 plus an additional $0.70 per mile driven. If the truck was driven 38 miles, what was the total amount Robert was charged?

A. $46.60

B. $52.90

C. $66.90

D. $86.50

1.9 6256e60f-a07b-4308-a386-94ecb7e20908 · Easy

The function g is defined by $gx = 4x - 6$. What is the value of g-7?

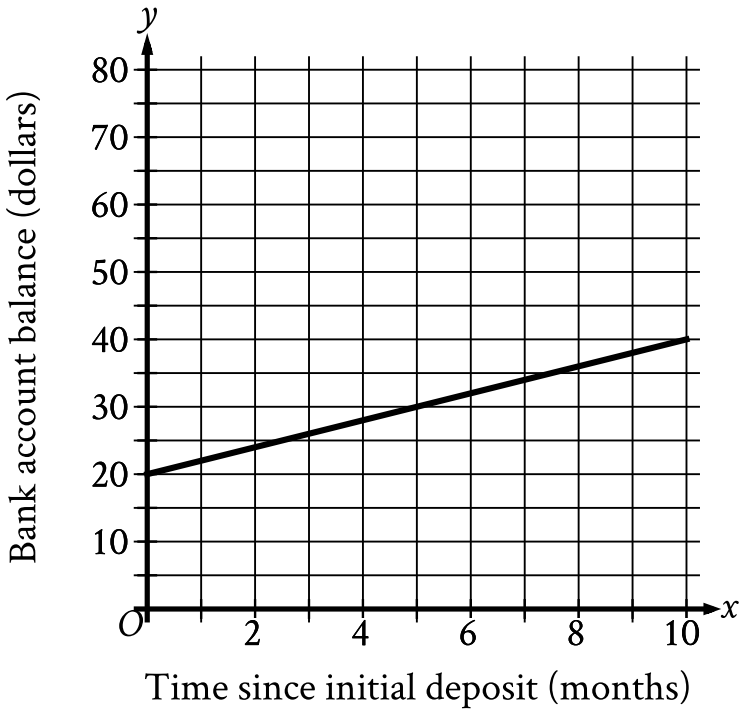
A. -34

B. -22

C. $-\frac{13}{4}$

D. $-\frac{1}{4}$

1.10 6ef925eb-e107-4c5c-a9ee-a6f89bf6d14e · Easy



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (0 comma 20)
  - (5 comma 30)
  - (10 comma 40)

A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

Student-produced response: _____

1.11 029386-DC · Easy

The graph of the function f is a line in the xy-plane. If the line has slope $\frac{3}{4}$ and $f(0)=3$, which of the following defines f?

A. $f(x)=\frac{3}{4}x-3$

B. $f(x)=\frac{3}{4}x+3$

C. $f(x)=4x-3$

D. $f(x)=4x+3$

1.12 84994068-0631-49ef-ae80-b174172efcd3 · Easy

$$d = 16 - \frac{x}{30}$$

The equation shown gives the estimated amount of diesel d, in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \le x \le 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

A. 0

B. 6

C. 14

D. 16

1.13 f1332d3b-a308-4640-95e0-cae936826d8d · Easy

The function f is defined by $f(x) = 25x + 30$. What is the value of fx when $x = 2$?

A. 50

B. 57

C. 80

D. 110

1.14 f7cecf09-36cf-4934-b397-765b7be1e77c · Easy

Josie purchased a pass that cost $\$ 19.50$ to tour a nature preserve. She also purchased 10 identical stickers to give to her nieces and nephews. The pass and the stickers cost $\$ 62.00$ total. What was the cost of one sticker?

A. $\$ 1.95$

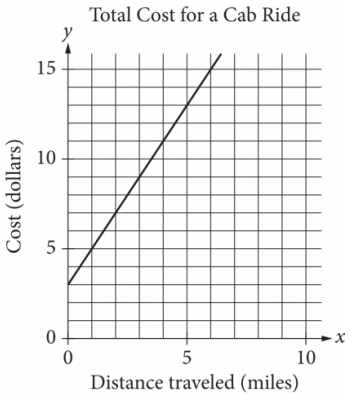
B. $\$ 4.25$

C. $\$ 6.20$

D. $\$ 8.15$

1.15 020326-DC · Easy

The line graphed in the xy-plane below models the total cost, in dollars, for a cab ride, y, in a certain city during nonpeak hours based on the number of miles traveled, x.



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

A. $2.00

B. $2.60

C. $3.00

D. $5.00

1.16 b5527b99-0bdb-46d2-9513-73a5fae75276 · Easy

The function f is defined by $f(x) = 3x - 8$. What is the value of $f7$?

A. 29

B. 13

C. -5

D. -29

1.17 b86303fd-698f-45e4-ba2a-97ad8f15a2d6 · Easy

A student council group is selling school posters for a fundraiser. They use the function $px = 5x - 220$ to determine their profit px, in dollars, for selling x school posters. In order to earn a profit of $\$ 900$, how many school posters must they sell?

Student-produced response: _____

1.18 2825ecca-ede4-4a29-bbb0-928be95b590b · Easy

$$fx = 7x + 1$$

The function gives the total number of people on a company retreat with x managers. What is the total number of people on a company retreat with 7 managers?

Student-produced response: _____

1.19 61063a9c-5dc4-47b2-858c-6c3052f4e804 · Easy

In the linear function h, $h(0) = 41$ and $h(1) = 40$. Which equation defines h?

A. $h(x) = -x + 41$

B. $h(x) = -x$

C. $h(x) = -41x$

D. $h(x) = -41$

1.20 08176-DC · Easy

If f is the function defined by $f(x) = \frac{2x-1}{3}$, what is the value of $f(5)$?

A. $\frac{4}{3}$

B. $\frac{7}{3}$

C. 3

D. 9

1.21 68357233-11ec-44c6-882e-eb8bb84450eb · Easy

$$fx = x + \frac{8}{11}$$

The function *f* is defined by the given equation. What is the value of *fx* when $x = \frac{3}{11}$?

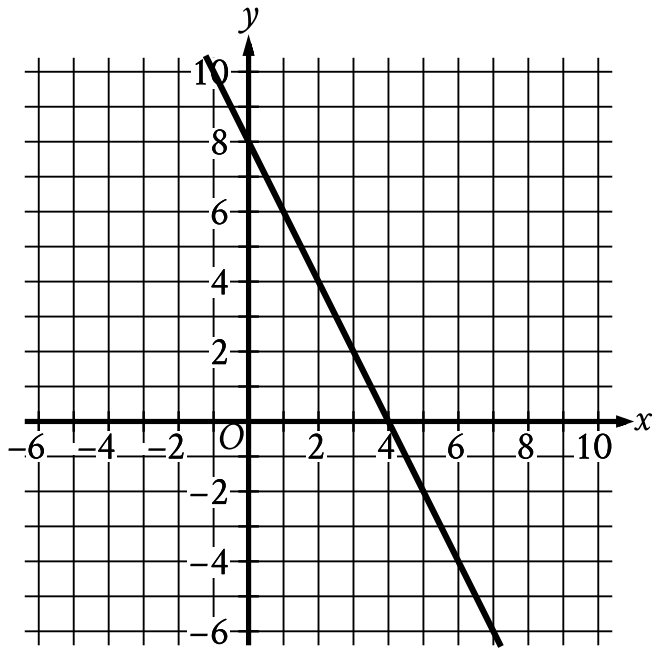
Student-produced response: _____

1.22 3e4d926d-83f1-4d65-87c3-c5889ee7bfa4 · Easy

The function *f* is defined by $fx = \frac{7}{10}x + 55$. What is the value of $f20$?

Student-produced response: _____

1.23 9db3fbf3-8d74-4521-aab7-7ad6bfc05c13 · Easy



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (0 comma 8)
  - (1 comma 6)

The graph of the linear function f is shown, where $y = fx$. Which equation defines function f?

A. $fx = -2x - 8$

B. $fx = -2x + 8$

C. $fx = 2x - 8$

D. $fx = 2x + 8$

1.24 d06b80b0-bba8-4218-bf79-f285777b8fbc · Easy

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters (cm), of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of 39 in this context?

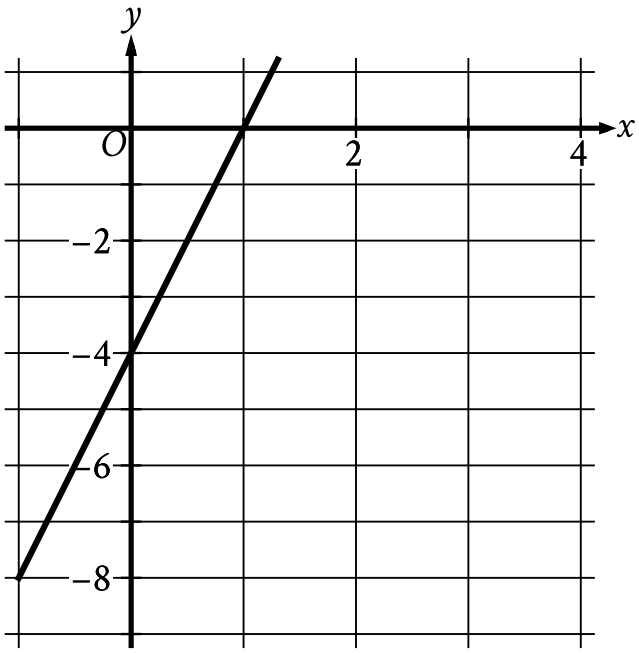
A. The estimated height, in cm, of the fluid at the start of the experiment

B. The estimated height, in cm, of the fluid at the end of the experiment

C. The estimated change in the height, in cm, of the fluid each day

D. The estimated number of days for all the fluid to evaporate

1.25 e35017e8-7f1b-42e1-a7bd-e47cb7628320 · Easy



- The line slants sharply up from left to right.
- The line passes through the following points:
  - (0 comma negative 4)
  - (1 comma 0)

The graph of the function f is shown, where $y = f(x)$. What is the y-intercept of the graph?

A. $0, -1$

B. $0, -4$

C. $0, 1$

D. $0, 4$

1.26 4c2cb31a-9ec5-40cc-863d-4baac298c63b · Easy

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d, in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker?

A. 30

B. 32

C. 60

D. 90

1.27 9dc23c3a-e1d2-402a-9e2a-4085bbba4933 · Easy

The function f is defined by $f(x) = 5x + 8$. For what value of x does $f(x) = 58$?

A. 10

B. 13

C. 50

D. 298

1.28 07aee0d5-9ae6-4612-bbac-cad96d3eb3dd · Easy

The total cost $f(x)$, in dollars, to lease a car for 36 months from a particular car dealership is given by $f(x) = 36x + 1{,}000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is $\$\ 400$?

A. $\$\ 13{,}400$

B. $\$\ 13{,}000$

C. $\$\ 15{,}400$

D. $\$\ 37{,}400$

1.29 de934a98-c3b0-4f0e-bccc-3bbfd3d0308d · Easy

Area (square feet)	Water (gallons)
2,520	4,536
3,780	6,804
5,040	9,072

The buildings of a shopping center are designed to allow water to drain from the roof into gutters on the sides of the buildings. The table shows the relationship between the area x, in square feet, of a roof and the amount of water fx, in gallons, drained from the roof into the gutters over a certain period of time. Which equation could define f?

A. $fx = 0.6x$

B. $fx = 1.8x$

C. $fx = 2{,}268x$

D. $fx = 4{,}536x$

1.30 ab05e5bc-239c-421b-a9ab-056838213249 · Easy

For the linear function f, the graph of $y = f(x)$ in the xy-plane has a slope of 7 and passes through the point $0, 5$. Which equation defines f?

A. $f(x) = 5x$

B. $f(x) = 35x$

C. $f(x) = 7x + 5$

D. $f(x) = 12x + 5$

1.31 2a9be9d6-4a3d-4f2e-92ee-f5423d312669 · Easy

The function f is defined by $fx = 92x + 3$. For what value of x does $fx = 63$?

A. 2

B. 5

C. 7

D. 30

1.32 bd9c2702-bba2-4f95-a9c2-d109cfd98cc1 · Easy

The function f is defined by $f(x) = 4x - 3$. What is the value of $f(10)$?

A. -30

B. 37

C. 40

D. 43

1.33 077b3b36-9acc-43fd-842b-44f85e95f12f · Easy

The function f is defined by the equation $fx = 7x + 2$. What is the value of fx when $x = 4$?

Student-produced response: _____

1.34 31182667-4cd9-4dea-af99-85f88f84bfa1 · Easy

A veterinarian recommends that each day a certain rabbit should eat 25 calories per pound of the rabbit's weight, plus an additional 11 calories. Which equation represents this situation, where c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit's weight is x pounds?

A. $c = 25x$

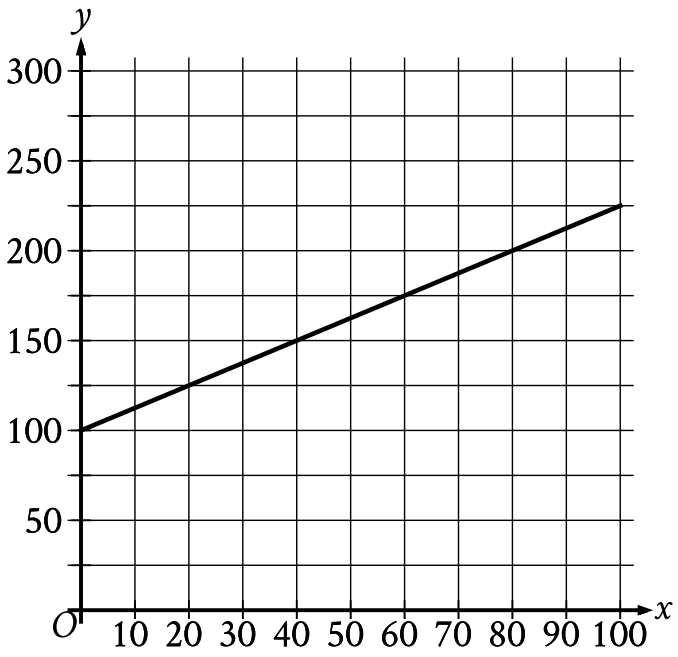
B. $c = 36x$

C. $c = 11x + 25$

D. $c = 25x + 11$

1.35 d37f3f28-86cf-4511-ab8e-f7639eafd76a · Easy

The cost y, in dollars, for a manufacturer to make x rings is represented by the line shown.



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (0 comma 100)
  - (60 comma 175)
  - (100 comma 225)

What is the cost, in dollars, for the manufacturer to make 60 rings?

A. 100

B. 125

C. 175

D. 225

1.36 bba90b53-4666-4123-babf-8f8dbe2f8a7f · Easy

Hana deposited a fixed amount into her bank account each month. The function $ft = 100 + 25t$ gives the amount, in dollars, in Hana's bank account after t monthly deposits. What is the best interpretation of 25 in this context?

A. With each monthly deposit, the amount in Hana's bank account increased by $\$ 25$.

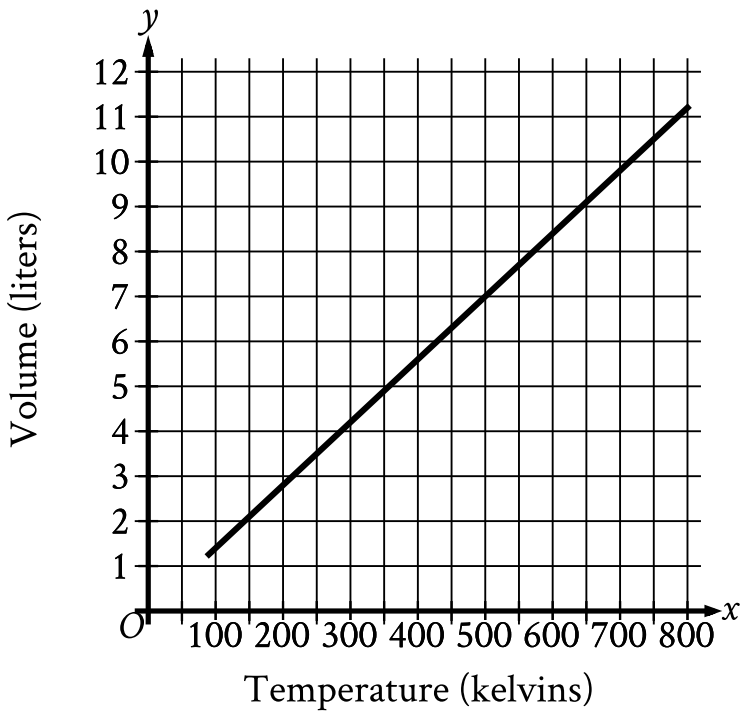
B. Before Hana made any monthly deposits, the amount in her bank account was $\$ 25$.

C. After 1 monthly deposit, the amount in Hana's bank account was $\$ 25$.

D. Hana made a total of 25 monthly deposits.

1.37 af90e68d-767f-4f40-82c5-58dbdc93234b · Easy

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume *y*, in liters, of the hydrogen when its temperature is *x* kelvins.



- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.3).
- The line passes through the following approximate points:
  - (90 comma 1.3)
  - (500 comma 7)

What is the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins?

A. 0

B. $\frac{7}{500}$

C. 7

D. $\frac{500}{7}$

1.38 545e9160-249b-4bdc-a5ec-c8c9e6fc08d6 · Easy

For the linear function f, the graph of $y = f(x)$ in the xy-plane has a slope of 39 and passes through the point $0, 0$. Which equation defines f?

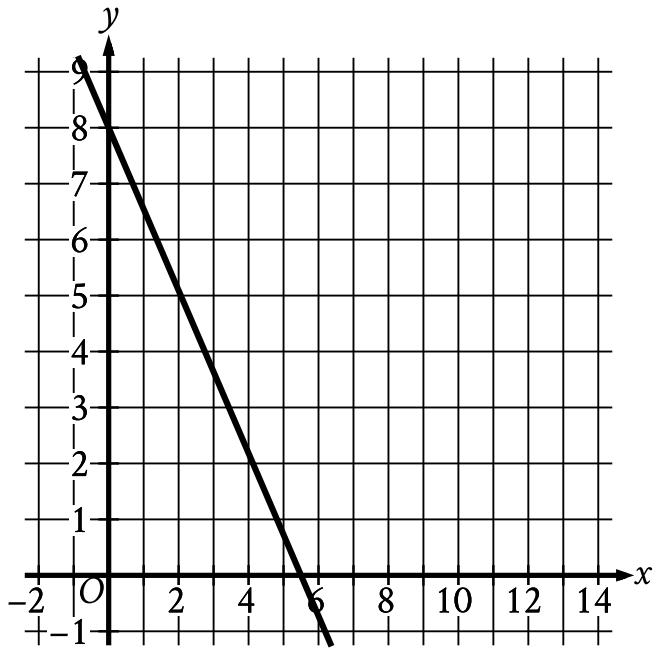
A. $f(x) = -39x$

B. $f(x) = \frac{1}{39}x$

C. $f(x) = x - 39$

D. $f(x) = 39x$

1.39 b057ae02-4dbf-4141-afa7-c0784af1f5b2 · Easy



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (0 comma 8)
  - (StartFraction 11 Over 2 EndFraction comma 0)

The graph of the linear function f is shown, where $y = f(x)$. What is the y-intercept of the graph of f?

A. $0, 0$

B. $0, -\frac{16}{11}$

C. $0, -8$

D. $0, 8$

1.40 08832-DC · Easy

Marisol drove 3 hours from City A to City B. The equation below estimates the distance d, in miles, Marisol traveled after driving for t hours.

$d = 45t$

Which of the following does 45 represent in the equation?

A. Marisol took 45 trips from City A to City B.

B. The distance between City A and City B is 45 miles.

C. Marisol drove at an average speed of about 45 miles per hour.

D. It took Marisol 45 hours to drive from City A to City B.

1.41 0bd2a2ec-0bbd-4cd3-a37f-8e35808487c2 · Easy

The function f is defined by $fx = 80 - 6x$. What is the value of $f7$?

A. 13

B. 38

C. 74

D. 81

1.42 b1f78aeb-1ccd-4387-9894-38cac3d88e77 · Easy

$$f(x) = x + b$$

For the linear function f, b is a constant. When $x = 0$, $f(x) = 30$. What is the value of b?

A. -30

B. $-\frac{1}{30}$

C. $\frac{1}{30}$

D. 30

1.43 a58102d0-5605-4f3b-91b2-44ba46a14ee3 · Easy

$$d = 16t$$

The given equation represents the distance d, in inches, where t represents the number of seconds since an object started moving. Which of the following is the best interpretation of 16 in this context?

A. The object moved a total of 16 inches.

B. The object moved a total of $16t$ inches.

C. The object is moving at a rate of 16 inches per second.

D. The object is moving at a rate of $\frac{1}{16}$ inches per second.

1.44 020395-DC · Easy

$P(t) = 250 + 10t$

The population of snow leopards in a certain area can be modeled by the function P defined above, where $P(t)$ is the population t years after 1990. Of the following, which is the best interpretation of the equation $P(30) = 550$?

A. The snow leopard population in this area is predicted to be 30 in the year 2020.

B. The snow leopard population in this area is predicted to be 30 in the year 2030.

C. The snow leopard population in this area is predicted to be 550 in the year 2020.

D. The snow leopard population in this area is predicted to be 550 in the year 2030.

1.45 05103-DC · Easy

x	$f(x)$
1	5
3	13
5	21

Some values of the linear function f are shown in the table above. Which of the following defines f ?

A. $f(x)=2x+3$

B. $f(x)=3x+2$

C. $f(x)=4x+1$

D. $f(x)=5x$

1.46 ad28edfc-9193-4278-8a8f-8f9b3a34406c · Easy

The function f is defined by the equation $fx = 100x + 2$. What is the value of fx when $x = 9$?

A. 111

B. 118

C. 900

D. 902

1.47 907da20e-eb78-4bed-85f6-ce91ac870f45 · Easy

The function h is defined by $hx = x + 200$. What is the value of $h50$?

A. 200

B. 250

C. 10,000

D. 50,200

1.48 015564-DC · Easy

The front of a roller-coaster car is at the bottom of a hill and is 15 feet above the ground. If the front of the roller-coaster car rises at a constant rate of 8 feet per second, which of the following equations gives the height h, in feet, of the front of the roller-coaster car s seconds after it starts up the hill?

A. $h = 8s + 15$

B. $h = 15s + \frac{335}{8}$

C. $h = 8s + \frac{335}{15}$

D. $h = 15s + 8$

1.49 025529-DC · Easy

$T = 1{,}000 + 18h$

In the equation above, T represents Brittany's total take-home pay, in dollars, for her first week of work, where h represents the number of hours she worked that week and 1,000 represents a sign-on bonus. If Brittany's total take-home pay was $1,576, for how many hours was Brittany paid for her first week of work?

A. 16

B. 32

C. 55

D. 88

1.50 0f9a48e3-fac3-492a-8104-05def2d974ce · Easy

Scientists collected fallen acorns that each housed a colony of the ant species *P. ohioensis* and analyzed each colony's structure. For any of these colonies, if the colony has x worker ants, the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y, in the colony. If one of these colonies has 58 worker ants, which of the following is closest to the predicted number of larvae in the colony?

A. 41

B. 61

C. 83

D. 190

1.51 baf1c184-aaa3-416b-ac7d-232d0c6ccc89 · Easy

For the linear function f, the graph of $y = fx$ in the xy-plane has a slope of $\frac{1}{4}$ and passes through the point $0, 5$. Which equation defines f?

A. $fx = \frac{1}{4}x + 5$

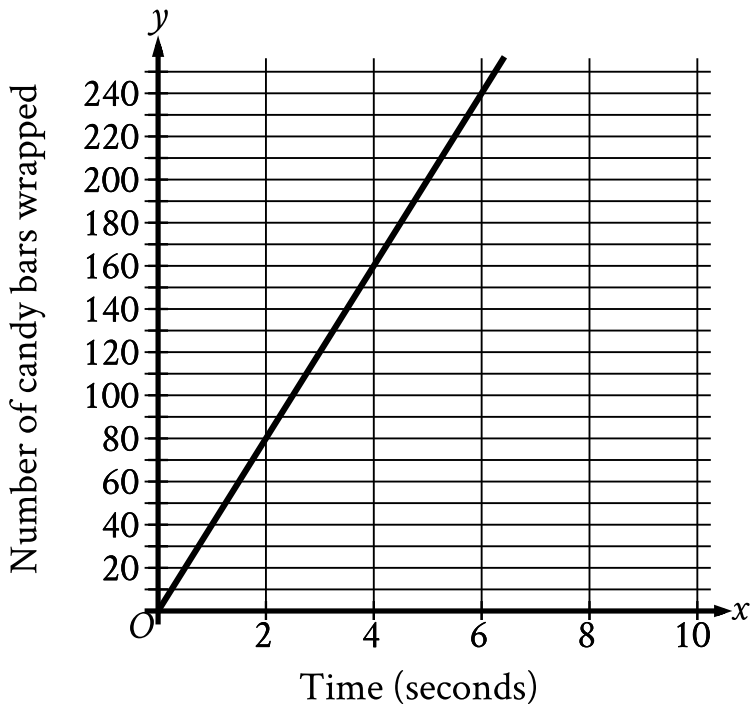
B. $fx = \frac{1}{4}x + \frac{1}{5}$

C. $f(x) = \frac{1}{4}x - \frac{5}{4}$

D. $fx = \frac{1}{4}x - 5$

1.52 a55ac55e-cf83-490d-9ed1-3e835edb8412 · Easy

The graph shown models the number of candy bars a certain machine wraps with a label in x seconds.



- The line slants sharply up from left to right.
- The line passes through the following points:
  - (0 comma 0)
  - (2 comma 80)
  - (4 comma 160)
  - (6 comma 240)

According to the graph, what is the estimated number of candy bars the machine wraps with a label per second?

A. 2

B. 40

C. 78

D. 80

1.53 1899a900-ac1b-40ac-a96b-d24684c6d46b · Easy

As part of a science project on evaporation, Amaya measured the height of a liquid in a container over a period of time. The function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. Which of the following is the best interpretation of 33 in this context?

A. The estimated height, in cm, of the liquid at the start of the project

B. The estimated height, in cm, of the liquid at the end of the project

C. The estimated change in the height, in cm, of the liquid each day

D. The estimated number of days for all of the liquid to evaporate

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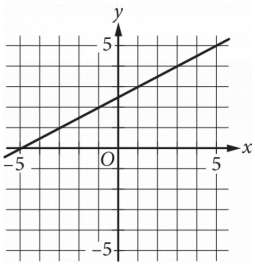
1.54 2e557c2f-0543-4d59-933b-131124398b14 · Easy

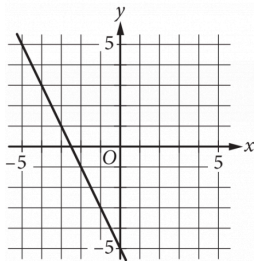
If $y = 5x + 10$, what is the value of y when $x = 8$?

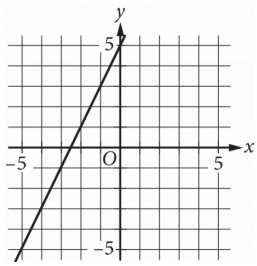
Student-produced response: _____

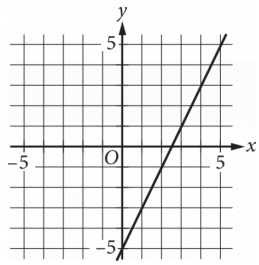
1.55 08449-DC · Easy

Which of the following is the graph of the equation $y = 2x - 5$ in the xy-plane?

A. 

B. 

C. 

D. 

1.56 b63fa866-0dfd-4de5-8017-16574107aa40 · Easy

The function g is defined by $gx = 10x + 8$. What is the value of gx when $x = 8$?

A. 0

B. 8

C. 10

D. 88

1.57 14b71de8-edf5-4dc4-a0fe-e77ab51f082f · Easy

$$gx = 11x + 4$$

For the given linear function g, which table shows three values of x and their corresponding values of gx?

A.

x	gx
-1	7
0	11
1	15

B.

x	gx
-1	-4
0	0
1	4

C.

x	gx
-1	-7
0	4
1	15

D.

x	gx
-1	-11
0	0

1	11

1.58 7fb9dc2e-7cc3-466f-b4fc-ac67568ce00b · Easy

The function f is defined by $fx = -3x + 60$. What is the value of fx when $x = -8$?

A. 49

B. 52

C. 57

D. 84

1.59 6c1cdc06-5eb3-4cb5-bca2-3f5aa2c8be38 · Easy

For a training program, Juan rides his bike at an average rate of 5.7 minutes per mile. Which function m models the number of minutes it will take Juan to ride x miles at this rate?

A. $mx = \frac{x}{5.7}$

B. $mx = x + 5.7$

C. $mx = x - 5.7$

D. $mx = 5.7x$

1.60 08177-DC · Easy

The function g is defined as $g(x) = 5x + a$, where a is a constant. If $g(4) = 31$, what is the value of a ?

A. 30

B. 22

C. 11

D. -23

1.61 79c4d360-5538-4b1d-992c-a822445814d8 · Easy

x	$f(x)$
0	29
1	32
2	35

For the linear function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines $f(x)$?

A. $f(x) = 3x + 29$

B. $f(x) = 29x + 32$

C. $f(x) = 35x + 29$

D. $f(x) = 32x + 35$

1.62 b4142eb8-7a46-4f1d-a34d-495ec7fb5693 · Easy

The function f is defined by $fx = \frac{1}{10}x - 2$. What is the y-intercept of the graph of $y = fx$ in the xy-plane?

A. $-2, 0$

B. $0, -2$

C. $0, \frac{1}{10}$

D. $\frac{1}{10}, 0$

1.63 27ae61df-4f9b-484e-84cb-efeecdfbdc47 · Easy

$$fx = 8x + 4$$

The function f gives the estimated height, in feet, of a willow tree x years after its height was first measured. Which statement is the best interpretation of 4 in this context?

A. The tree will be measured each year for 4 years.

B. The tree is estimated to grow to a maximum height of 4 feet.

C. The estimated height of the tree increased by 4 feet each year.

D. The estimated height of the tree was 4 feet when it was first measured.

1.64 ce1f679e-3746-457e-9ae8-995927d4a604 · Easy

$$f(x) = 14 + 4x$$

The function f represents the total cost, in dollars, of attending an arcade when x games are played. How many games can be played for a total cost of $\$ 58$?

Student-produced response: _____

1.65 49a43146-5d4a-4013-b46e-e0aa9a32fd1d · Easy

Sean rents a tent at a cost of $\$ 11$ per day plus a onetime insurance fee of $\$ 10$. Which equation represents the total cost c, in dollars, to rent the tent with insurance for d days?

A. $c = 11d + 10$

B. $c = 10d + 11$

C. $c = 11d + 10$

D. $c = 10d + 11$

1.66 ec93b358-6d54-4c6a-9e2b-fac8a57da030 · Easy

For the linear function f, the graph of $y = f(x)$ in the xy-plane has a slope of 2 and has a y-intercept at $0, -5$. Which equation defines f?

A. $f(x) = \frac{1}{2}x - 5$

B. $f(x) = -\frac{1}{2}x - 5$

C. $f(x) = -2x - 5$

D. $f(x) = 2x - 5$

1.67 a27bd477-cd97-482c-8d5d-ec57f4486fff · Easy

The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. Which of the following is the best interpretation of 9 in this context?

A. Tavon will keep the vine plant for 9 months.

B. The vine plant is expected to grow 9 inches each month.

C. The vine plant is expected to grow to a maximum length of 9 inches.

D. The estimated length of the vine plant was 9 inches when Tavon purchased it.

1.68 bfbc2147-51fa-4780-a273-d0cf88011d1c · Easy

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

A. 8

B. 9

C. 64

D. 80

1.69 027789-DC · Easy

The function g is defined by $g(x) = -x + 8$. What is the value of $g(0)$?

A. -8

B. 0

C. 4

D. 8

1.70 b7b056ed-d7cc-4901-9a1b-f261a1455bee · Easy

For the function f, the graph of $y = f(x)$ in the xy-plane has a slope of 3 and passes through the point $0, -8$. Which equation defines f?

A. $f(x) = 3x$

B. $f(x) = 3x - 8$

C. $f(x) = 3x + 5$

D. $f(x) = 3x + 11$

1.71 ddfbf736-4c36-47b1-8e69-a9b951ed1842 · Easy

$$s = 40 + 3t$$

The equation gives the speed s, in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

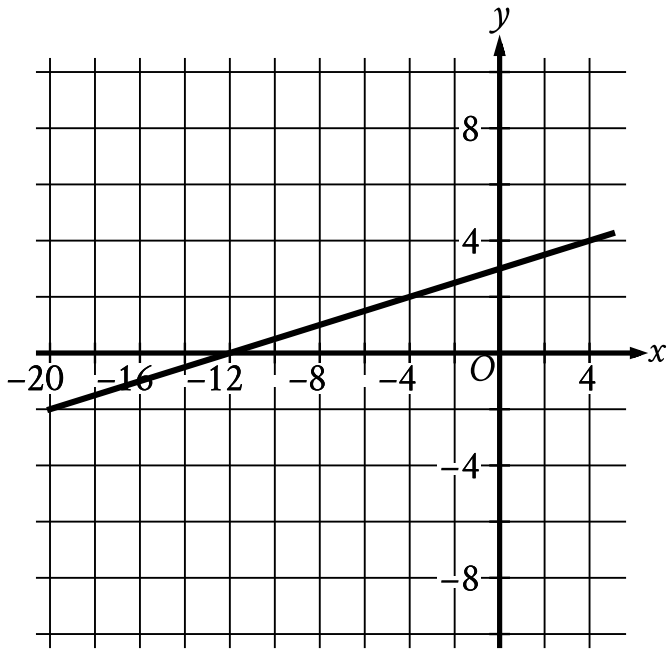
A. 40

B. 43

C. 45

D. 55

1.72 72165a25-0c05-4d08-b575-bdacc4a3d7fe · Easy



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 12 comma 0)
  - (negative 4 comma 2)

The graph of the linear function *f* is shown, where $y = fx$. What is the *x*-intercept of the graph of *f*?

A. $-12, 0$

B. $0, 0$

C. $\frac{1}{4}, 0$

D. $12, 0$

1.73 fa2330cf-1cd4-47f3-bf5c-5adbf6482253 · Easy

A computer company pays customers for trading in old computers. The payment p, in dollars, is the original retail price of a computer minus $\$ 150$ for each year since the computer was released. Which equation represents this situation for a computer with an original retail price of $\$ 690$ when it was released t years ago, where t is a whole number and $1 \leq t \leq 4$?

A. $p = 150 - 690t$

B. $p = 690 - 150t$

C. $t = 150 - 690p$

D. $t = 690 - 150p$

1.74 cdf32e4b-02c1-4140-951c-3af80dec3e9f · Easy

The length, y, of a white whale was 162 centimeters cm when it was born and increased an average of 4.8 cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in cm, of the whale?

A. $y = 162x$

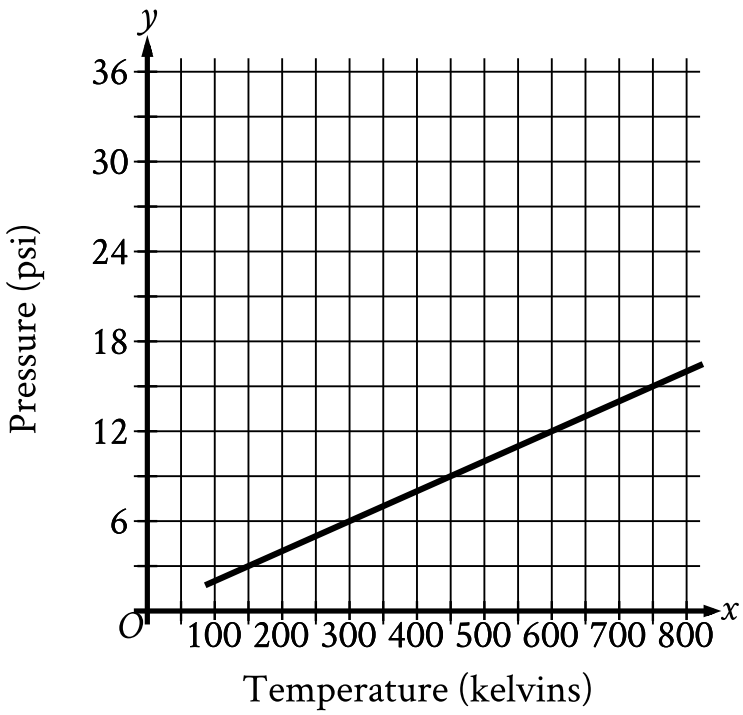
B. $y = 162x + 162$

C. $y = 4.8x + 4.8$

D. $y = 4.8x + 162$

1.75 b5fb74ca-504a-48fd-8613-f6d71ec02d98 · Easy

Argon is placed inside a container with a constant volume. The graph shows the estimated pressure y, in pounds per square inch psi, of the argon when its temperature is x kelvins.



- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.8).
- The line passes through the following points:
  - (300 comma 6)
  - (450 comma 9)
  - (600 comma 12)

What is the estimated pressure of the argon, in psi, when the temperature is 600 kelvins?

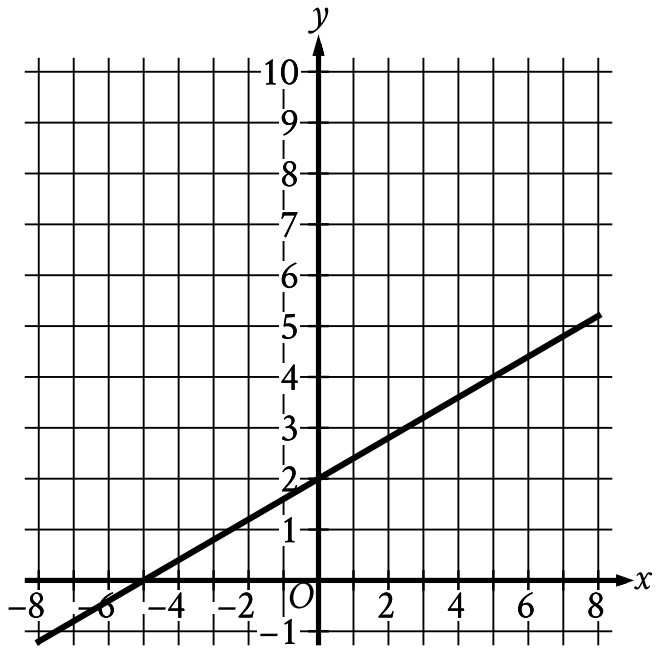
A. 6

B. 12

C. 300

D. 600

1.76 c300a207-fa7b-4c25-876b-37c0579a9bff · Easy



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 5 comma 0)
  - (0 comma 2)

The graph of the linear function *f* is shown. What is the *y*-intercept of the graph of $y = fx$?

A. $-5, 0$

B. $2, 0$

C. $0, 2$

D. $0, -5$

1.77 017100-DC · Easy

On January 1, 2015, a city's minimum hourly wage was $9.25. It will increase by $0.50 on the first day of the year for the next 5 years. Which of the following functions best models the minimum hourly wage, in dollars, x years after January 1, 2015, where $x = 1, 2, 3, 4, 5$?

A. $f(x) = 9.25 - 0.50x$

B. $f(x) = 9.25x - 0.50$

C. $f(x) = 9.25 + 0.50x$

D. $f(x) = 9.25x + 0.50$

1.78 f2255d61-7225-4d95-b076-0e4c0d8b749d · Easy

The function *f* is defined by $fx = \frac{1}{2}x + 6$. What is the value of *f*4?

A. 20

B. 12

C. 10

D. 5

1.79 3f8715bd-7d20-480b-a809-bd3b31a8df26 · Easy

The function *f* is defined by $fx = 6.2x$. What is the value of fx when $x = 10$?

A. 72

B. 62

C. 17.2

D. 16.2

1.80 d3019424-6572-4d26-990d-bc85bde91bda · Easy

The function h is defined by $hx = 3x - 7$. What is the value of h-2?

A. -13

B. -10

C. 10

D. 13

1.81 1a57825e-d5a2-43a1-ad8c-be3b91a796de · Easy

For the linear function f, $f0 = 17$ and $f1 = 17$. Which equation defines f?

A. $fx = \frac{1}{17}$

B. $fx = 1$

C. $fx = 17$

D. $fx = 34$

1.82 0f91f69e-0e9c-452b-aef8-15e6c57f7327 · Easy

To repair a refrigerator, a technician charges $\$ 60$ per hour for labor plus $\$ 120$ for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

A. $fx = x + 120$

B. $fx = 60x$

C. $fx = 60x + 120$

D. $fx = 60x - 120$

1.83 cc329e0c-3593-4615-8425-0a35606bd546 · Medium

The function f is defined by $f(x) = \frac{x+15}{5}$, and $f(a) = 10$, where a is a constant. What is the value of a?

A. 5

B. 10

C. 35

D. 65

1.84 b4b8b378-61b2-456e-8bc3-884fa80f6a40 · Medium

$$h(x) = x + b$$

For the linear function h, b is a constant and $h(0) = 45$. What is the value of b?

Student-produced response: _____

1.85 e65207ac-6145-49ce-842e-ade8831b09cf · Medium

$$g(m) = -0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

A. 0.05

B. 12.1

C. 20

D. 242.0
